

Anisotropic Casimir stress on rectangular flat T^3 and the absence of a free-field $H_t/H_p = 13/12$ attractor

Brendon Boyd^{1,*}

¹*Independent Researcher, Burswood, Western Australia, Australia[†]*

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Compact spatial topology can source anisotropic vacuum stress. For a free massless scalar on a rectangular flat three-torus we validate a renormalized lattice Casimir energy density and directional pressures (cutoff convergence, cubic isotropy, thermodynamic pressure derivatives, conformal trace, and dimensional scaling). Coupled as a small radiation-like perturbation to biaxial Bianchi-I expansion on a positive de Sitter background, free-field backreaction produces only *transient* passages near the historically proposed ratio $H_t/H_p = 13/12$, with no quasi-plateau or late-time attractor in a systematic shape–amplitude search. Persistent anisotropic expansion therefore cannot be attributed to free Casimir stress alone. We do not claim a resolution of the Hubble tension, a derivation of the galactic acceleration scale, or a completed cubic cosmology.

I. INTRODUCTION

Compact flat three-manifolds admit mode spectra that generically produce direction-dependent vacuum expectation values of the stress tensor [1–3]. That fact is distinct from any claim that a fixed numerical ratio of directional Hubble rates is a free parameter-free prediction of topology.

A historically attractive packaging of the latter idea is the cycle-counting assignment $H_t/H_p = 13/12$ together with the numerical identification $H_t = 72.97 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from a Planck anchor. Those statements are *not* conclusions of the present paper. Here we isolate two controlled questions that *can* be answered with a public lattice engine and a constrained biaxial integrator (gate package CBR-001):

1. Does a free massless scalar on rectangular T^3 produce a validated anisotropic Casimir stress that depends on the shape moduli?
2. When that stress is allowed to backreact on biaxial expansion, does $H_t/H_p = 13/12$ appear as a free-field attractor, a quasi-plateau, or only as a transient crossing (if at all)?

The answers are: yes to shape-dependent anisotropic stress; and only transient crossings of $13/12$, with every valid candidate returning to isotropy by late expansion, for the free-field source tested.

Companion work on present-epoch scale matching and weak-field matching invariants [4] records independent no-go results for a different geometric packaging; the Casimir calculation below does not rely on those withdrawn shortcuts and does not restore them.

II. RECTANGULAR LATTICE CASIMIR STRESS

A. Mode sum and pressures

For a massless scalar with periodic lengths (L_1, L_2, L_3) , the renormalized energy density used in CBR-001 is the rectangular-periodic Epstein-lattice result developed in Ref. [3]:

$$\rho_{\text{Cas}} = -\frac{\hbar c}{2\pi^2} \sum_{\mathbf{n} \in \mathbb{Z}^3 \setminus \{0\}} \frac{1}{[(n_1 L_1)^2 + (n_2 L_2)^2 + (n_3 L_3)^2]^2}. \quad (1)$$

Directional pressures follow thermodynamically,

$$p_i = -\frac{1}{L_j L_k} \frac{\partial E_{\text{Cas}}}{\partial L_i}, \quad E_{\text{Cas}} = L_1 L_2 L_3 \rho_{\text{Cas}}. \quad (2)$$

In code and tables we report dimensionless lattice units with the overall $\hbar c$ factor restored only when matching cosmological amplitudes. Figure 1 recalls the rectangular fundamental domain (not an embedded doughnut T^2).

B. Stage-1 validation (cubic point)

At the cube $L_1 = L_2 = L_3 = 1$ the cutoff sequence extrapolates to

$$\rho = -0.83753589, \quad p_i = \rho/3 = -0.27917863 \quad (3)$$

in lattice units, with conformal-trace residual $|\rho - \sum_i p_i| \sim 10^{-15}$ at $N = 120$ modes per edge, finite-difference pressure errors $\sim 5 \times 10^{-10}$, and cubic isotropy at machine precision (Table I). The Stage-1 package reports STATUS: PASS.

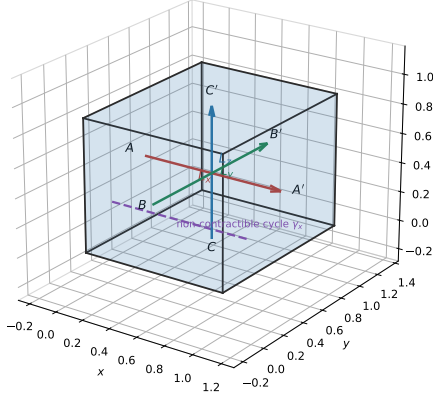
C. Stage-2 biaxial shape scan

Let $r = L_t/L_p$ with fixed mean scale in the scan convention of CBR-001 Stage 2. The stress $\rho(r)$, $p_p(r)$, $p_t(r)$

* brendon.boyd@itsm-cosmology.org
ORCID: 0009-0007-4177-2612

[†] GitHub: github.com/brendohxd/ITSM-...
Zenodo: [doi:10.5281/zenodo.18808348](https://doi.org/10.5281/zenodo.18808348)
Gate: Analysis/Casimir/CBR-001/

Flat T^3 fundamental domain (rectangular): opposite faces identified
 $T^3 \simeq \mathbb{R}^3/\Lambda$ (not the doughnut surface $T^2 \subset \mathbb{R}^3$)



Identifications: $A \sim A'$, $B \sim B'$, $C \sim C'$. Cubic: $L_x = L_y = L_z$. Rectangular: independent L_i .

FIG. 1. Rectangular flat T^3 fundamental domain with independent edge lengths L_p and L_t (biaxial case $L_2 = L_3 = L_t$). This is the geometry of the lattice sum, not a Planck-safe cubic cosmology and not an embedded solid torus.

TABLE I. Cubic Stage-1 diagnostics (lattice units, $L_i = 1$). Extrapolated ρ uses an $O(1/N^2)$ cutoff fit; validation residuals are at mode cutoff $N = 120$.

Quantity	Value
ρ (extrap.)	-0.83753589
p_i (extrap.)	-0.27917863
max FD pressure rel. error	4.8×10^{-10}
conformal-trace residual	$\sim 10^{-15}$
cube isotropy error	$\sim 10^{-16}$

and the anisotropy $\Delta p \equiv p_t - p_p$ are smooth functions of shape; Δp changes sign across the cube $r = 1$ (Fig. 2). Representative points (lattice units) include

$$r = 1 : \quad \rho = -0.8375, \quad \Delta p \simeq 0, \quad (4)$$

$$r \simeq 1.26 : \quad \rho = -0.4702, \quad \Delta p \simeq +0.247. \quad (5)$$

Compact *shape-dependent* anisotropic stress is therefore a concrete, validated free-field mechanism. It is *not* a universal cycle-counting coefficient independent of the mode sum.

III. BIAxIAL FREE-FIELD BACKREACTION

A. Geometry and equations

For the biaxial Bianchi-I line element

$$ds^2 = -dt^2 + a_p^2 dx^2 + a_t^2 (dy^2 + dz^2), \quad (6)$$

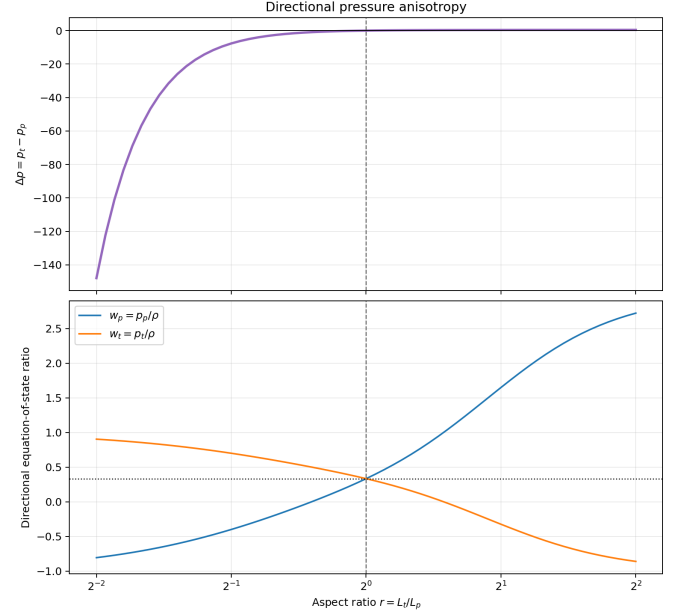


FIG. 2. Stage-2 biaxial anisotropy $\Delta p = p_t - p_p$ versus shape ratio $r = L_t/L_p$. The cubic point $r = 1$ is isotropic; anisotropy is shape moduli dependent.

define

$$H = \frac{H_p + 2H_t}{3}, \quad \delta = H_t - H_p, \quad r = \frac{a_t}{a_p}. \quad (7)$$

The shear system is the biaxial specialization of Bianchi-I [5], with $\kappa = 8\pi G$,

$$\dot{\delta} + 3H\delta = \kappa(p_t - p_p), \quad \dot{r} = r\delta, \quad (8)$$

and Hamiltonian constraint

$$3H^2 - \frac{\delta^2}{3} = \kappa\rho_{\text{total}}. \quad (9)$$

Stage 3A treats the verified Casimir tensor as a small radiation-like perturbation ($\propto a^{-4}$) on a positive de Sitter background and integrates in $N = \ln a$. The dimensionless amplitude ϵ absorbs $\hbar c/L_\star^4$ relative to the de Sitter density scale (CBR-001 Stage 3A convention). It validates the engine (perturbative limit, constraint and continuity residuals) but does *not* by itself establish any preferred Hubble ratio.

B. The Stage-3B ratio test

Let

$$x = \frac{\delta}{H}, \quad q = \frac{H_t}{H_p} = \frac{1 + x/3}{1 - 2x/3}. \quad (10)$$

The historical target $q = 13/12$ corresponds to $x_\star = 3/38$. The small-source fixed-background profile $x(N) =$

TABLE II. Stage-3B classification at threshold amplitudes for the 13/12 search. “Regime” labels the peak Casimir energy fraction (perturbative / marginal / nonperturbative).

r_0	class	ϵ_*	regime
1.01	no crossing	—	—
1.05	no crossing	—	—
1.10	transient	2.55	nonpert.
1.25	transient	1.34	nonpert.
1.50	transient	0.676	nonpert.
2.00	transient	0.278	nonpert.
3.00	transient	0.089	marginal
4.00	invalid	—	—

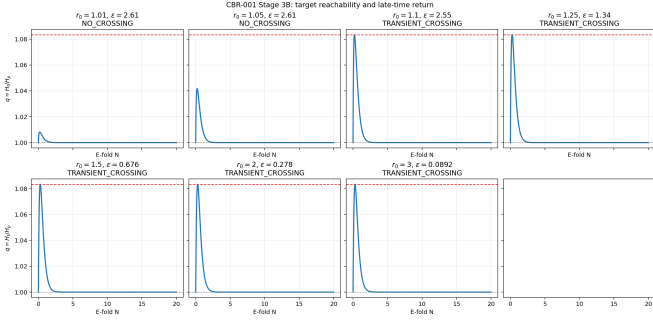


FIG. 3. Representative Stage-3B trajectories of $q = H_t/H_p$ versus N . The horizontal line marks 13/12. Passages, when present, are brief; late evolution returns to isotropy.

$S(e^{-3N} - e^{-4N})$ peaks at $N = \ln(4/3)$ with maximum $(27/256)S$; merely reaching x_* already requires $S \geq 128/171 \simeq 0.7485$, outside the strict small-perturbation regime.

A nonlinear root search over initial shapes $r_0 \in \{1.01, 1.05, 1.1, 1.25, 1.5, 2, 3, 4\}$ and amplitudes spanning six decades finds target-reaching thresholds for five of eight shapes. Classification vocabulary:

- **NO_CROSSING**: peak q stays below 13/12;
- **TRANSIENT_CROSSING**: touch or brief crossing;
- **QUASI_PLATEAU**: dwell ≥ 1 e-fold within 1% without attractor criteria;
- **ATTRACTOR**: dwell ≥ 5 e-folds within 1%, nearby trajectories converge, and no subsequent decay toward $q = 1$.

Results: zero attractors, zero quasi-plateaus, five transient crossings, two no-crossings, one invalid (out-of-domain) shape (Table II; Fig. 3). Dwell times within 1% of the target are only ~ 0.27 – 0.30 e-folds. Every valid candidate returns to $q \rightarrow 1$ by $N = 20$, consistent with an a^{-4} free-field source becoming negligible. Nearby-trajectory stability tests confirm decay toward isotropy; none satisfy attractor criteria.

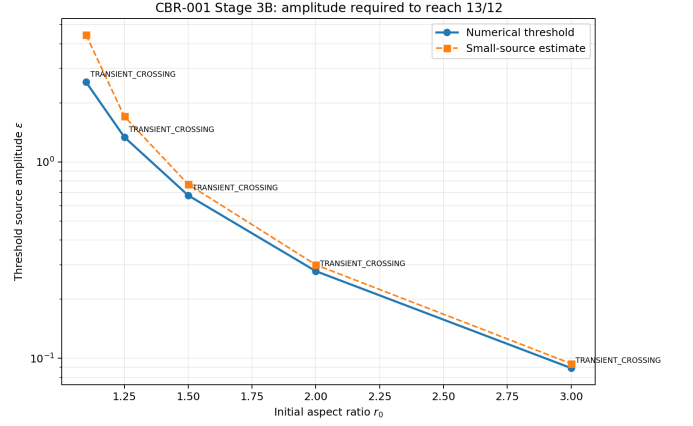


FIG. 4. Stage-3B threshold amplitudes $\epsilon_*(r_0)$ for first touch of $q = 13/12$. Near-cubic shapes require large free-field amplitudes; no threshold defines an attractor.

C. Kinematic warning

Because $d \ln r / dN = x$, a constant $q \neq 1$ requires continuous evolution of the shape ratio; it is not a fixed-torus property. A persistent anisotropic state would have to be a *driven* solution with an additional constitutive stress, not a free Casimir attractor. For a prescribed $x(N)$, the required anisotropic pressure is

$$\Pi_{\text{req}} = \frac{H^2}{\kappa} \left[\frac{dx}{dN} + x \left(3 + \frac{d \ln H}{dN} \right) \right]. \quad (11)$$

In de Sitter at fixed $x = 3/38$ this is $\Pi_{\text{req}} = 9H^2/(38\kappa)$. The free Casimir anisotropy scales as

$$\Pi_{\text{Cas}} \propto a^{-4} r^{8/3} [\hat{p}_t(r) - \hat{p}_p(r)] \quad (12)$$

and cannot maintain Π_{req} once the source redshifts away. Any future positive claim of a persistent 13/12-class ratio must derive that extra stress (open programme CBR-002) without inserting the target into a coefficient by hand.

IV. HISTORICAL PACKAGING (EXPLICITLY NOT RESTORED)

Cycle counting that assigns a fixed fraction 1/12 per compact direction and promotes $H_t/H_p = 13/12$ to a free-field prediction, together with the numerical map $H_p = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1} \Rightarrow H_t = 72.97 \text{ km s}^{-1} \text{ Mpc}^{-1}$, is a packaging of the *question* this paper tests—not a result we endorse. The lattice stress is shape dependent; free-field backreaction does not lock the ratio; and CMB topology bounds on cubic E_1 cells [6, 7] are a separate observational sector that this rectangular free-field simulation does not claim to satisfy.

We likewise do not claim that rectangular Casimir stress derives a galactic acceleration scale, fixes weak-field matching coefficients, or resolves the Hubble tension [8, 9].

V. LIMITATIONS AND OPEN ROUTES

- Free massless scalar only; interacting or multi-field sectors untested.
- Rectangular moduli; twisted flat spaces (E_2 , E_3) not scanned.
- Positive de Sitter testbed with Casimir as a small perturbation—not a Casimir-dominated universe.
- Persistent anisotropy requires a derived driven or memory stress (CBR-002), not free-field packaging.
- CMB compatibility of any particular (L_i) is not established here.

These limitations are research boundaries, not research bans: driven constitutive stress, alternate topologies, and richer field content remain open under an honest claim ladder.

VI. CONCLUSIONS

A free massless scalar on rectangular flat T^3 yields a validated, shape-dependent anisotropic Casimir stress. When that stress backreacts as a small a^{-4} source on biaxial de Sitter expansion, the ratio $H_t/H_p = 13/12$ appears at most as a brief transient at large free-field amplitudes, never as a quasi-plateau or attractor in the search reported here. Free Casimir stress alone therefore cannot sustain the historically proposed anisotropic expansion, nor justify a free-field map to $H_t = 72.97 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Public code and stage summaries under `Analysis/Casimir/CBR-001/` reproduce

the Stage-1–3B pipeline. Frozen SHA-256 anchors for the numerical tables used here are recorded in the paper repository file `CBR001_CHECKSUMS.md` (data and code availability for this draft).

ACKNOWLEDGMENTS

Computations use the repository environment `itsm_env` and the CBR-001 gate package. Claim hygiene for related geometric packaging is recorded in companion note [4].

Appendix A: Reproducibility anchors

Table III lists SHA-256 digests of the CBR-001 CSV/JSON artifacts underlying the Stage-1–3B numbers in the main text (branch `recovery/v12-core-architecture`, freeze date 2026-08-01). Re-running the Stage scripts after intentional code changes will alter digests; any published revision must update this table.

TABLE III. SHA-256 of CBR-001 artifacts used in this draft.

Artifact	SHA-256 (prefix)
<code>cbr001_stage1.csv</code>	9E6874226FAA...
<code>cbr001_stage2_scan.csv</code>	9257F3A67CB5...
<code>cbr001_stage3b_thresholds.csv</code>	370B02480FF0...
<code>cbr001_stage3b_summary.json</code>	4AD9223CE32C...

Full digests and reproduction commands appear in `papers/P2-Rectangular-T3-Casimir/CBR001_CHECKSUMS.md`.

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